Stems: Forms and Function

Stem Functions

- Stems have the following 4 functions or jobs:
  - Support
  - Conduction
  - Growth
  - Storage

Support

- Stems support the plant's leaves, flowers, and fruits.

Conduction

- Like roots, stems contain vascular tissue (xylem and phloem).
- Xylem conducts water and minerals.
- Phloem conducts the sugars created during photosynthesis.

Growth

- During the growing season, cell division and elongation are occurring in both the apical and lateral meristems of dicots.
- This results in plant growth.
- The same things are occurring in the intercalary meristems of monocots.

Storage

- Stems are sometimes modified to store food and (or) water.
- Plants that store large amounts of water in their stems (cactus) or leaves (succulents).

Cactus

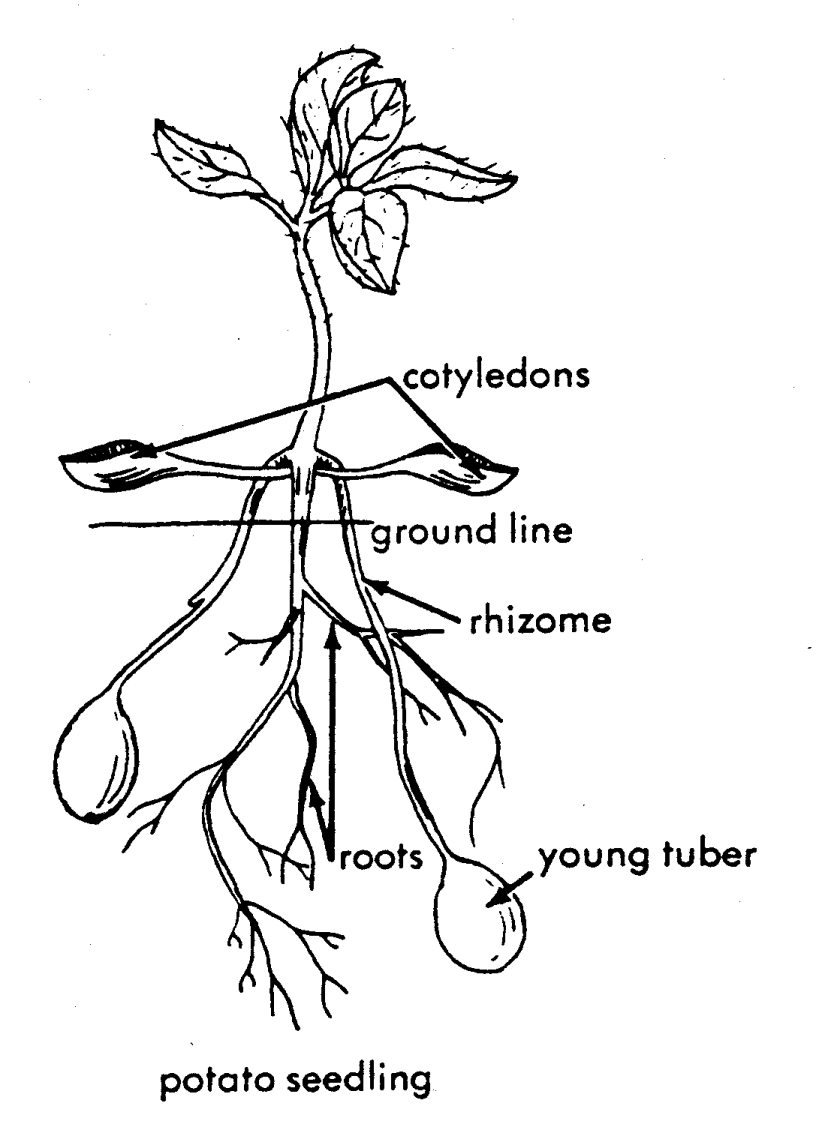
- Cacti are examples of succulent plants that store water in their stems.
- The spines are modified leaves.



Tuber

- A potato is a special type of underground stem called a tuber.
- Tubers store excess carbohydrates manufactured during photosynthesis.
- Don’t confuse tubers with tuberous roots. Tuberous roots are root modifications, tubers are modified stems.

The Potato is a Tuber





mature tuber

Woody Dicot Stem Morphology

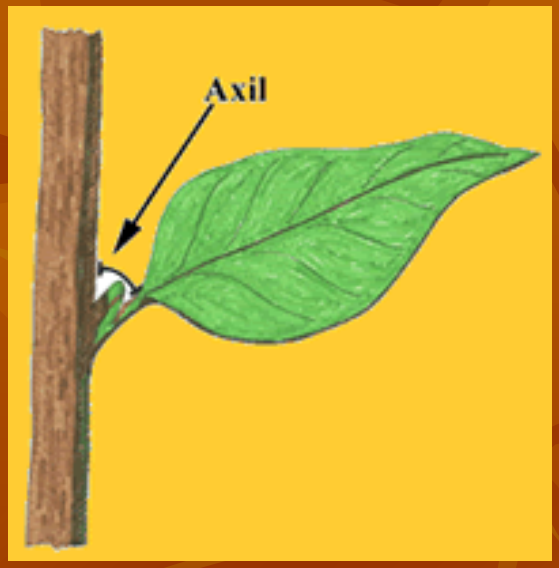
- **Nodes**: places on stems where leaves or buds are located.
- **Leaf axil**: where a leaf petiole attaches to a stem (the angle created by the intersection of the petiole and stem).
- **Internodes**: the sections of a stem located between the nodes.

Woody Dicot Stem Morphology





Node



Stem Morphology, continued

- **Bud**: a structure on a stem containing a dormant apical meristem.
  - Buds are usually, but not always, covered by protective scales.
  - Can be terminal or lateral in position.
  - Come in 3 basic types:
    - Vegetative
    - Flower
    - Mixed


bud scales


naked bud

Woody Dicot Stem Morphology



Lilac mixed buds breaking dormancy.

Mixed buds of Lilac are opening up to reveal both stems/leaves and flowers.

Stem Morphology, continued

- **Terminal bud scale scars**: mark the location of last seasons terminal bud.
- **Leaf scars**: scars marking the location of last seasons leaves.
- **Vascular bundle scars**: found inside leaf scars, these scars mark the location of the vascular bundles that passed from the stem, through the leaf petiole, and into the leaf blade.

Woody Dicot Stem Morphology



Stem Morphology, continued

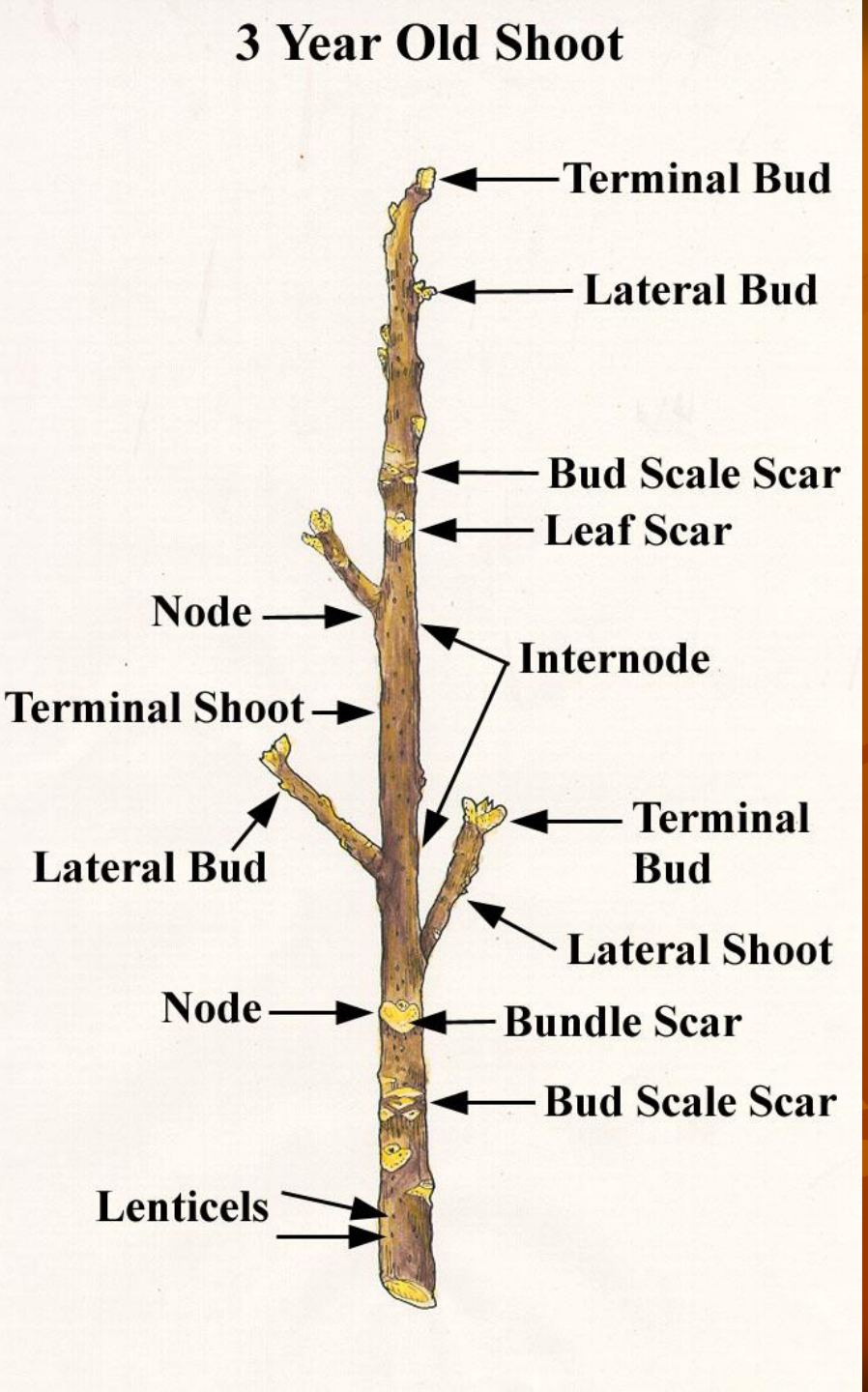
- **Lenticels**: dot-like structures on stems that consist of loosely packed cells that may play a role in gas exchange between the atmosphere and stem. Not all stems have lenticels.

Woody Dicot Stem Morphology

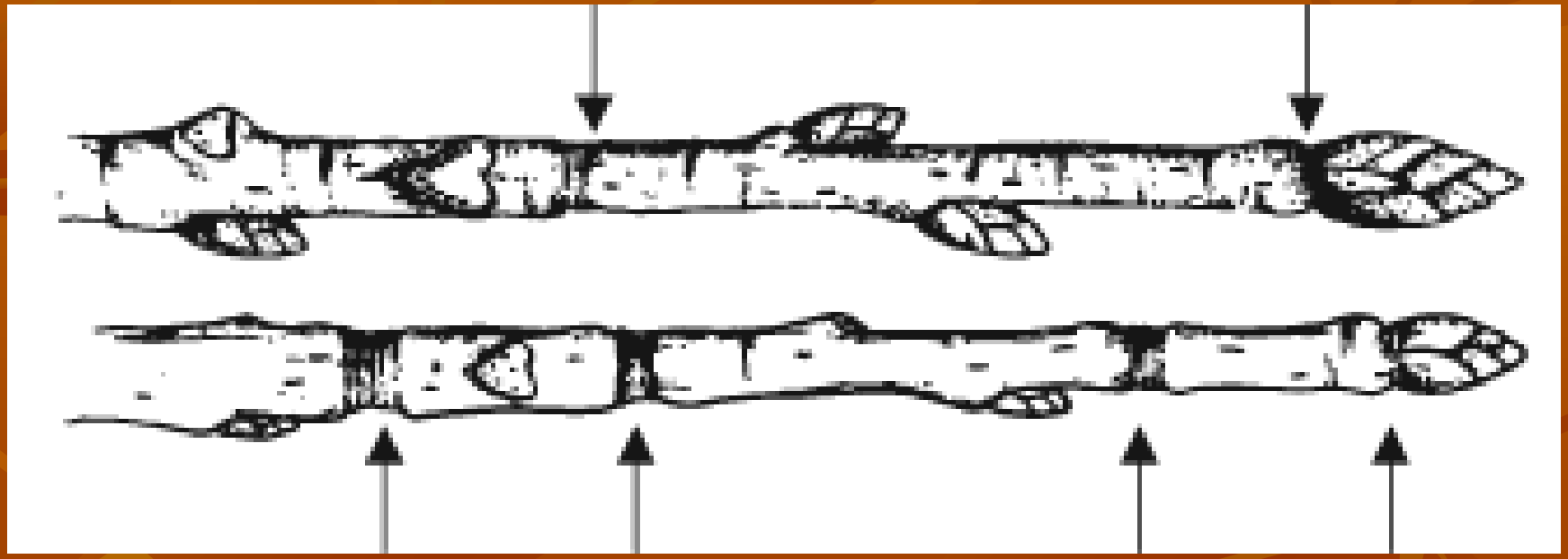


Catalpa stem with leaf scar, lateral bud and lenticels.

Woody Dicot Stem Morphology



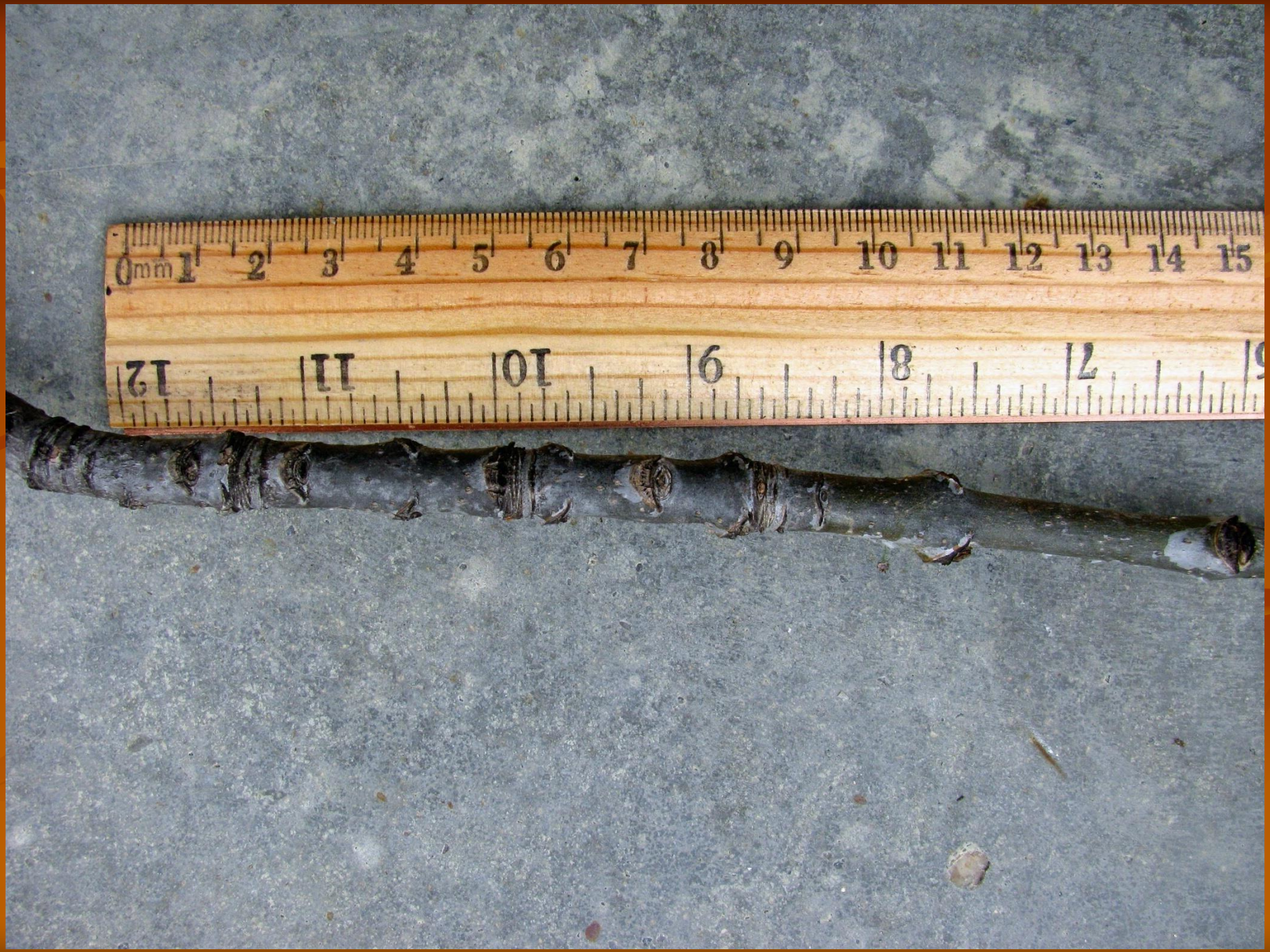
Woody Dicot Stem Morphology



Arrows mark the position of the terminal bud scale scars


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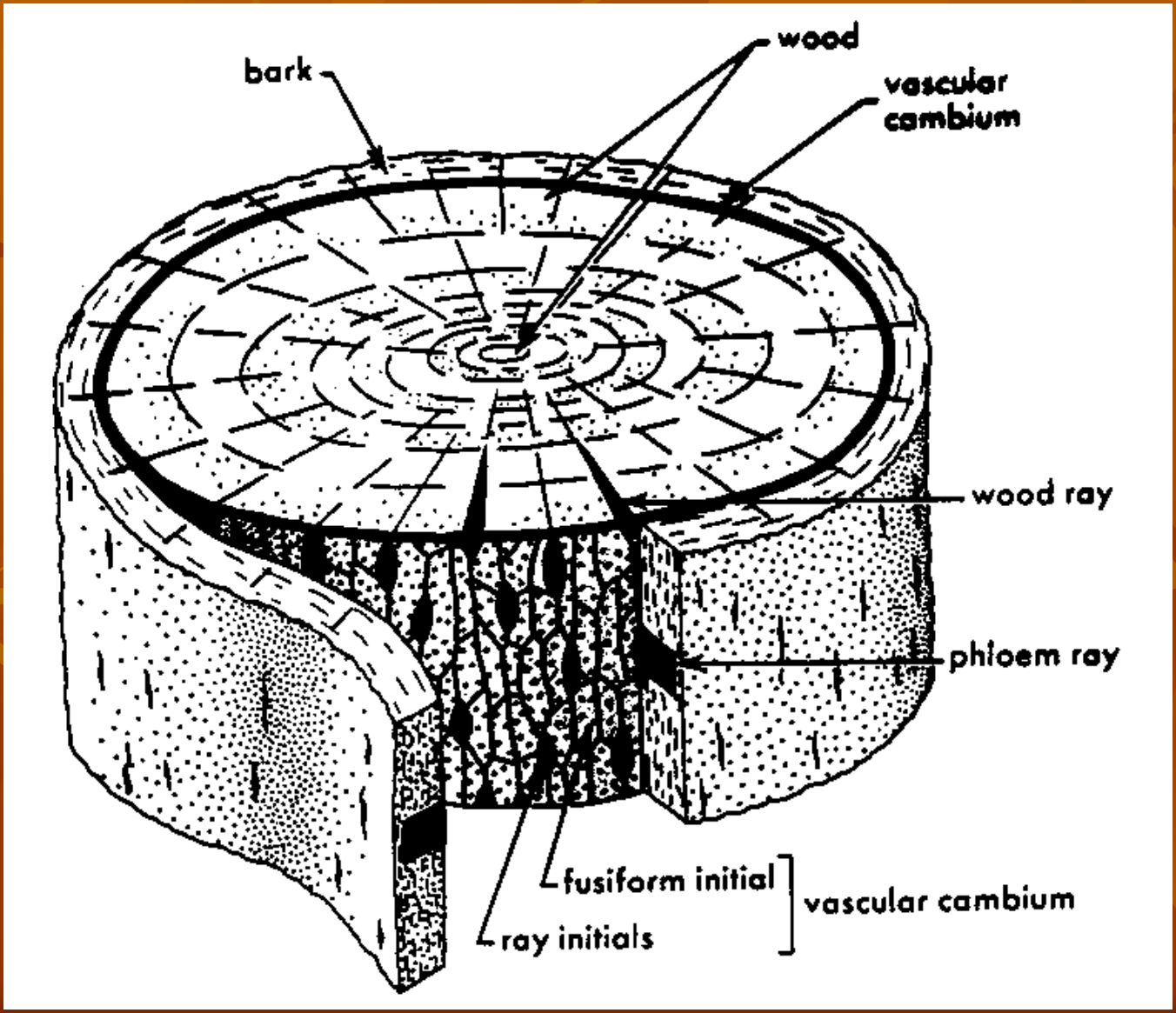
Woody Dicot Stem Anatomy

- The cells in the stem of a woody, dicot plant are oriented in 2 different directions.
- Xylem and phloem are *vertically* arranged tissues.
- The vascular cambium produces bundles of *horizontally* arranged cells called vascular rays or just rays for short.
- Rays pass through both xylem and phloem.

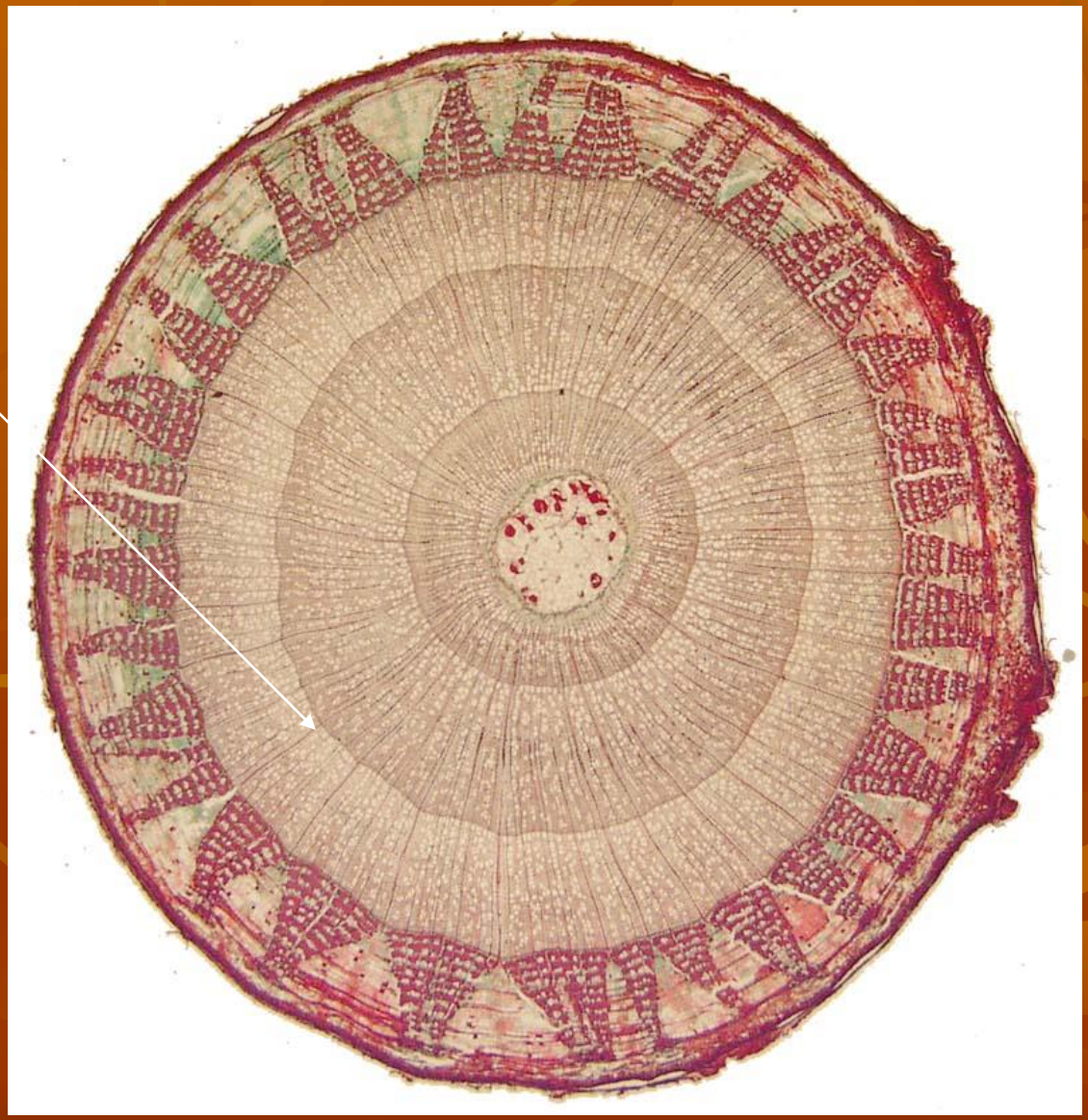
Vascular Rays

- Rays allow for the *lateral movement* of water, dissolved minerals, sugars, and waste products within the stem.

Stem Anatomy



Woody Dicot Stem Cross-Section



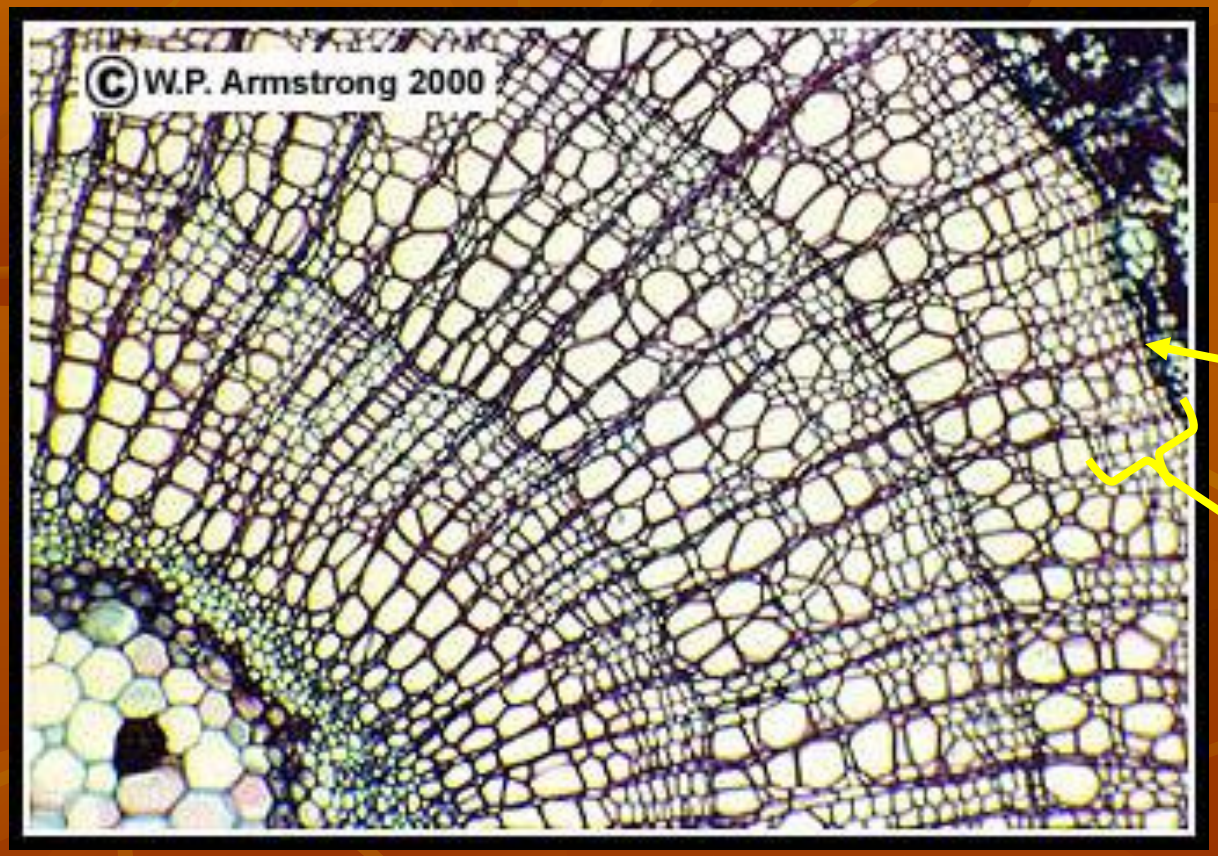
Stem Anatomy Terminology

- **Wood:** xylem tissue
- **Springwood:** large, thin-walled xylem cells produced in the spring of the year when water is usually in abundant supply.
- **Summerwood:** smaller, thicker-walled xylem cells produced during the summer when water availability is more limited.

Stem Anatomy Terminology

- **Annual Ring:** one annual growth ring represents the amount of springwood and summerwood that has occurred in a single growing season.
- Trees growing in the tropics, where the temperature is constant and rainfall occurs daily, do not produce annual growth rings.

Springwood-Summerwood



Cross-sectional view of a 3-year old woody, dicot stem

Vascular Cambium

Summerwood yr 3

Springwood yr 1

Summerwood yr 1

Springwood yr 2

Summerwood yr 2

Springwood yr 3

Heartwood

- Heartwood: As xylem ages it becomes plugged with waste products.
- When this happens the xylem (wood) becomes stained a dark color.
- Heartwood is not functional xylem. It no longer conducts water.
- It does, however, help strengthen the trunk of a tree.

Sapwood

- Sapwood: xylem that is actively conducting water.
- Lighter in color than heartwood.
- Surrounds the heartwood.

Heartwood/Sapwood



Maple Syrup

- In late summer-early fall starches are stored in stem and root sapwood.
- The following year, in early spring when the temperature is right, these starches are converted to sugars.
- The watery, sugar-filled sap begins flowing upwards.
- Trees are tapped and the sap is collected in buckets.
- Later, the water is boiled-off to make maple syrup.





Stem Modifications

- We have already mentioned tubers as being one type of stem modification that is used to store carbohydrates (food).
- In the following slides we will examine other types of stem modifications.

Rhizomes

- An underground stem.
- Usually grows horizontally.
- Capable of producing new shoots and roots.





Stolons

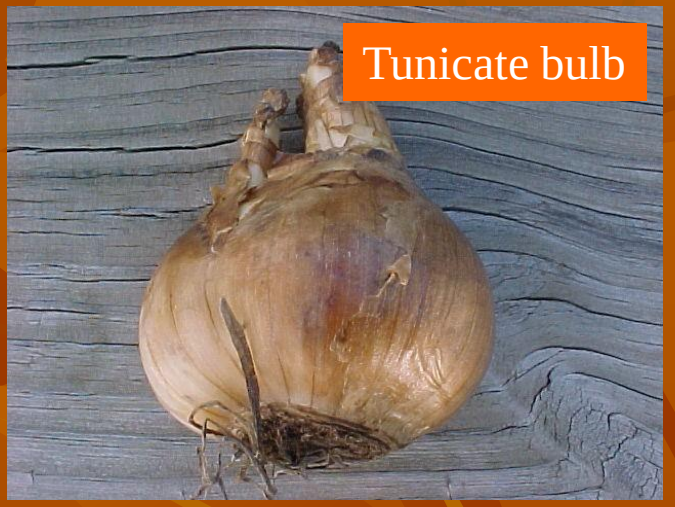
- Aboveground stems that “run” across the soil surface.
- Horizontally growing.
- Produce new shoots and roots.



Strawberry stolons

Bulbs

- Short stem with fleshy leaves that function as food storage organs during dormancy
- Two different kinds:
  - Tunicate bulbs have a papery outer covering. An onion and garlic are examples.
  - Scaly bulbs lack a papery outer covering. Lily bulbs are an example.





Parts of a Bulb

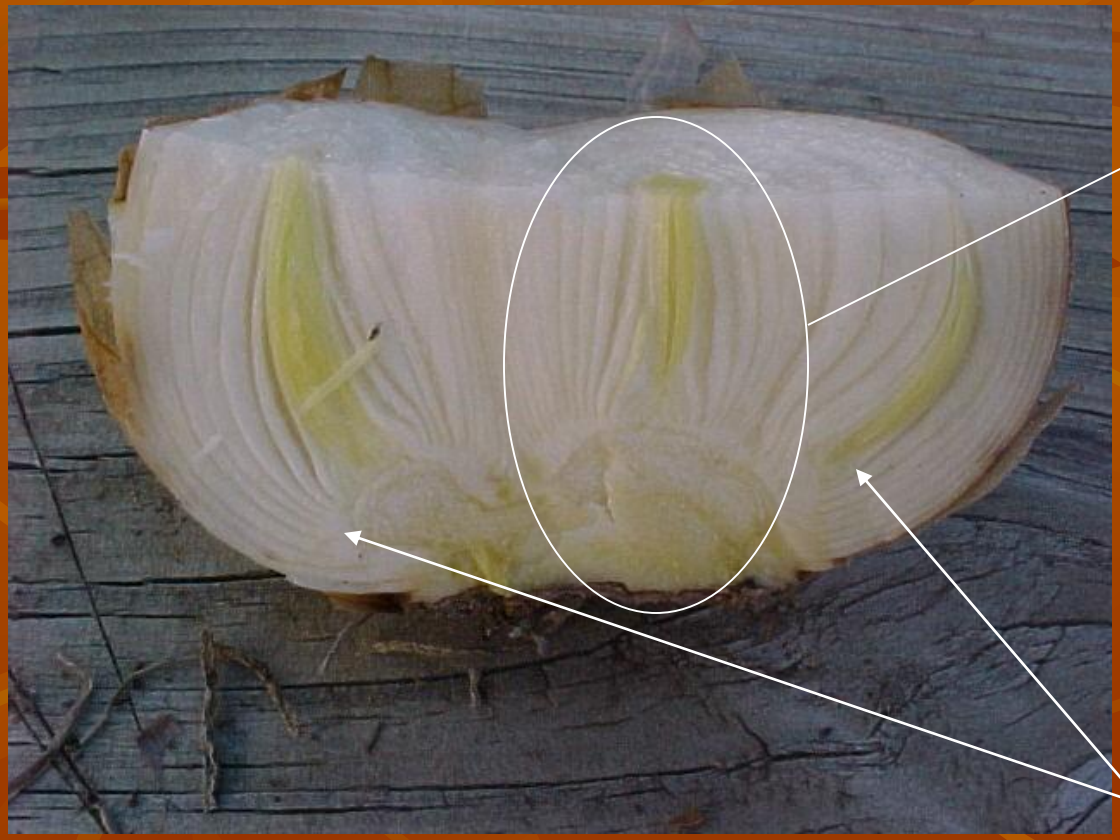


Scales: modified leaves that store carbohydrates

Apical bud: sits on top of the basal plate. Will eventually form a shoot bearing leaves and flowers

Basal plate: modified stem

Offset Bulbs



The End